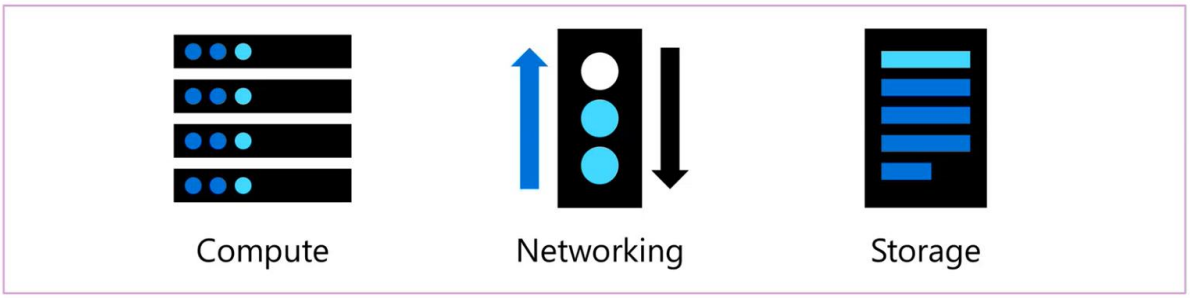
Cloud Computing 101

What is Cloud Computing

Cloud computing is the delivery of computing services such as servers, storage, databases, software, networking and analytics over the internet, enabling faster innovation, flexible resources and economies of scale.



Types of Cloud Computing (Deployment Models)

Private Cloud

- Organizations create a cloud environment in their data center
- Organizations are responsible for operating the services they provide
- Does not provide access to users outside of the organization

Public Cloud

- Owned by cloud services or hosting providers
- Provides resources and services to multiple organizations and users
- Accessed via secure network connection (typically over the internet)

Hybrid Cloud

Combines public and private clouds to allow applications to run in the most appropriate location.

Other Cloud Models

Community Cloud: Is shared by organizations with similar needs

Multi-Clouds: Using more than one cloud provider

Cloud model comparison

Public cloud

- No capital expenditures to scale up.
- Applications can be quickly provisioned and deprovisioned.
- Organizations pay only for what they use.

Private cloud

- Hardware must be purchased for start-up and maintenance.
- Organizations have complete control over resources and security.
- Organizations are responsible for hardware maintenance and updates.

Hybrid cloud

- Provides the most flexibility.
- Organizations determine where to run their applications.
- Organizations control security, compliance, or legal requirements.

Compare CapEx vs. OpEx

Capital expenditure (CapEx)

- The upfront spending of money on physical infrastructure.
- Costs from CapEx have a value that reduces over time.

Operational expenditure (OpEx)

- Spend on products and services as needed, pay-as-you-go.
- Get billed immediately.

Consumption-based model

Cloud service providers operate on a consumption-based model, which means that end users only pay for the resources that they use.

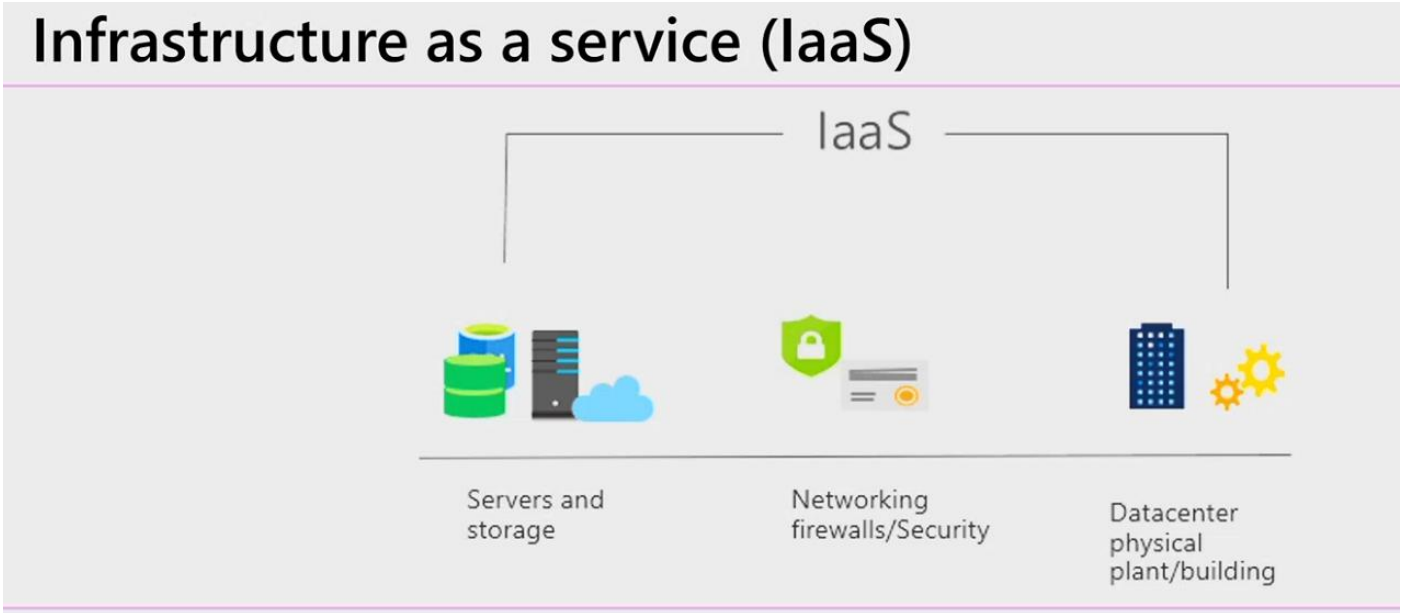
- Better cost prediction.
- Prices for individual resources and services are provided.
- Billing is based on actual usage.

Cloud Benefits

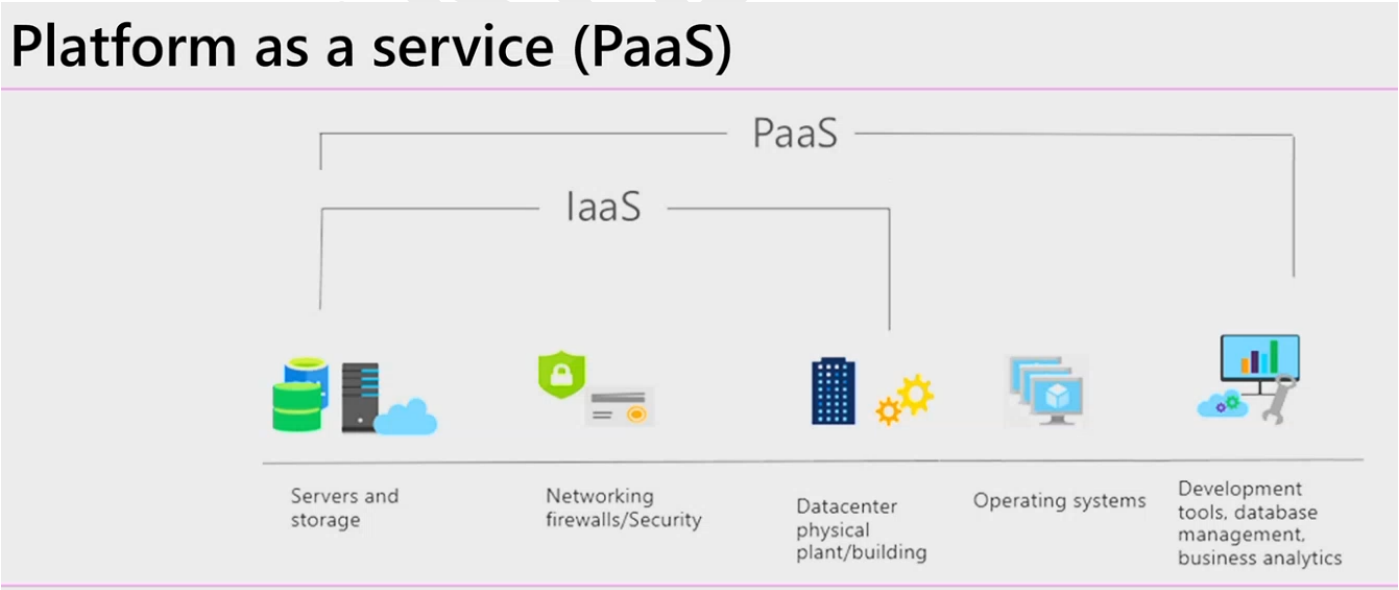
- **High availability:** accessible most of the time from any location
- **Elasticity:** handles changes automatically and dynamically.
- **Scalability:** easily allocate or deallocate resources as needs grow or shrink.
- **Reliability:** designed to recover quickly from failures and continue operating without data loss.
- **Predictability:** usage and costs can be accurately estimated and planned
- **Security:** implement strong physical and digital security measures to protect data and systems.

- **Governance:** ensures resources are compliant with policies and regulations
- **Manageability:** resources are easily monitored, configured, and maintained.

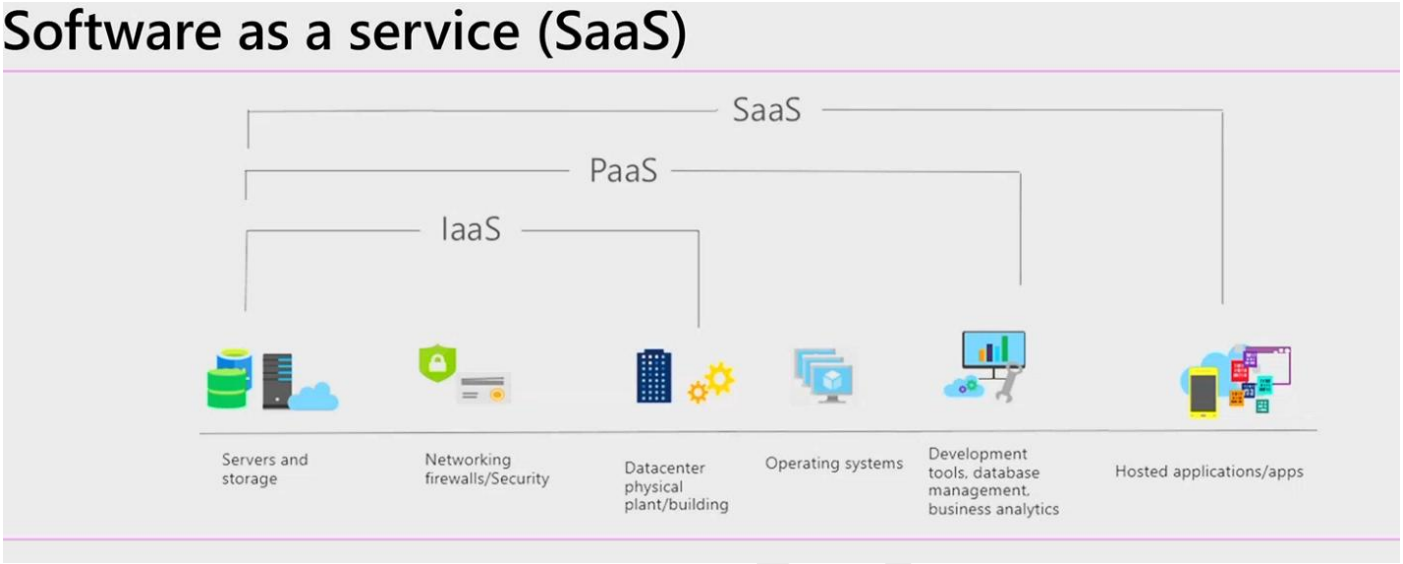
Cloud Service Types



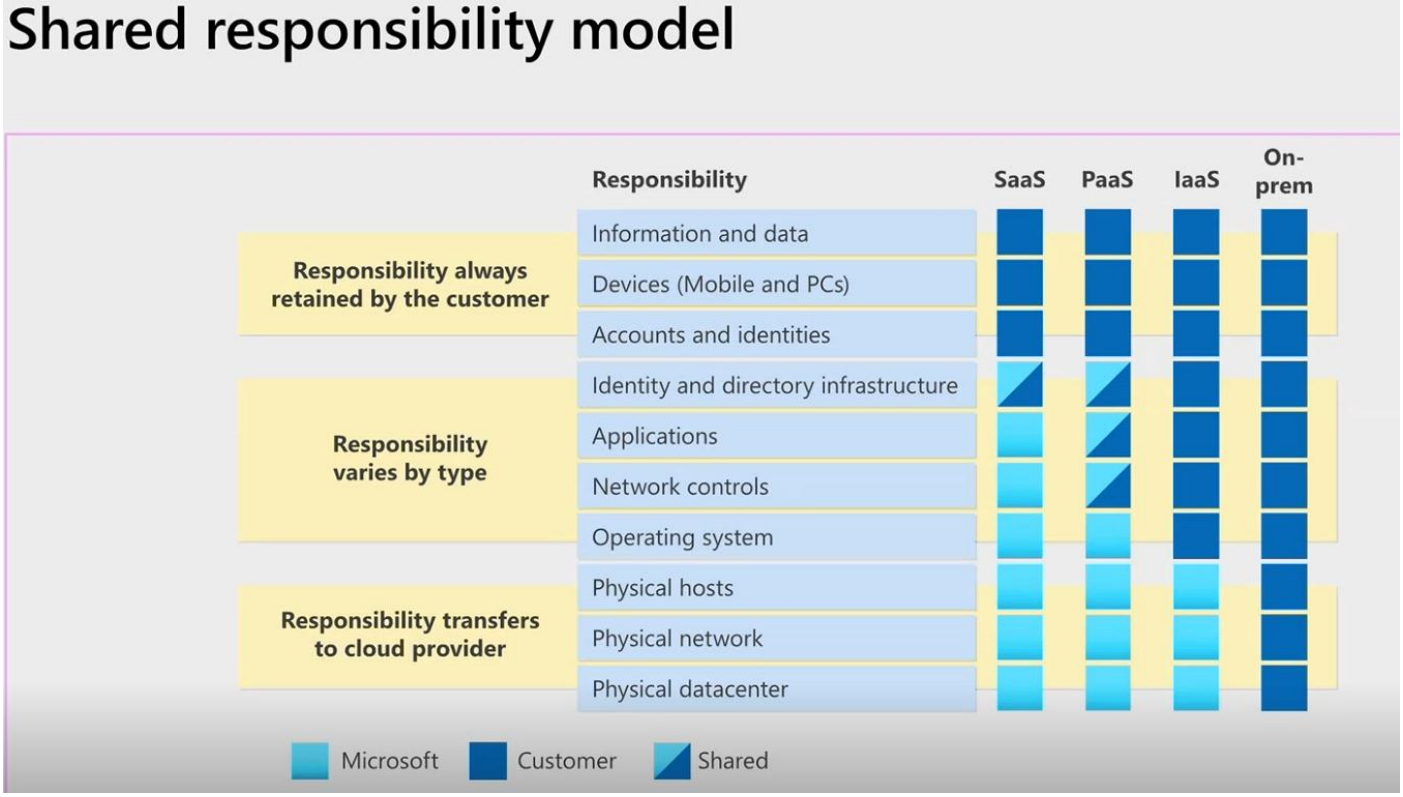
IaaS builds a pay-as-you-go IT infrastructure by renting servers, virtual machines, storage networks and operating systems from a cloud provider.



PaaS provides an environment for building, testing and deploying software applications without focusing on managing the underlying infrastructure.



Users connect to and use cloud-based apps over the internet e.g. Microsoft office 365, emails and calendars.



Cloud service comparison

IaaS

- The most flexible cloud service.
- You configure and manage the hardware for your application.

PaaS

- Focus on application development.
- Platform management is handled by the cloud provider.

SaaS

- Pay-as-you-go pricing model.
- Users pay for the software they use on a subscription model.

DIVINE UZOR